7 Assembling and using an agent platform

This chapter covers

- Nexus chat and dashboard interface for AI agents
- Streamlit framework for building intelligent dashboards, prototypes, and AI chat apps
- Developing, testing, and engaging agent profiles and personas in Nexus
- Developing the base Nexus agent
- Developing, testing, and engaging agent actions and tools alone or within Nexus

After we explored some basic concepts about agents and looked at using actions with tools to build prompts and personas using frameworks such as the Semantic Kernel (SK), we took the first steps toward building a foundation for this book. That foundation is called Nexus, an agent platform designed to be simple to learn, easy to explore, and powerful enough to build your agent systems.

7.1 Introducing Nexus, not just another agent platform

There are more than 100 AI platforms and toolkits for consuming and developing large language model (LLM) applications, ranging from toolkits such as SK or LangChain to complete platforms such as AutoGen and CrewAI. This makes it difficult to decide which platform is well suited to building your own AI agents.

Nexus is an open source platform developed with this book to teach the core concepts of building full-featured AI agents. In this chapter, we'll